

Is it Ella's birthday?

Ella's birthday is July 28th. This program takes a day and a month and uses logical operators to check whether it's Ella's birthday.



1 Check for the birthday

Create variables for a day and a month. Use the "and" logical operator to check whether it is July 28th.

```
>>> day = 28
>>> month = 7
>>> day == 28 and month == 7
True
```

The "and" operator checks to see if both conditions are true

Remember to use two equals signs

It's Ella's birthday!

2 Not the birthday detector

You can reverse the answer using the "not" logical operator. You will get the answer "True" on every day, except for Ella's birthday.

```
>>> day = 28
>>> month = 7
>>> not (day == 28 and month == 7)
False
```

This character is used to make code go over two lines

It's Ella's birthday, so the answer is "False"

3 Birthday or New Year's Day?

Use the "or" logical operator to check whether it's Ella's birthday or New Year's Day. Use brackets to combine the correct days and months.

```
>>> day = 28
>>> month = 7
>>> (day == 28 and month == 7) \
    or (day == 1 and month == 1)
True
```

Checks for the 28th of July

The answer will be "True" if it's Ella's birthday or New Year's Day

Strings

Two strings can be compared using the "==" operator or the "!=" operator. Strings have to match exactly to get a "True" output.

```
>>> dog = 'Woof woof'
>>> dog == 'Woof woof'
True
>>> dog == 'woof woof'
False
>>> dog == 'Woof woof '
False
```

The strings match exactly, so the answer is "True"

The strings don't match because there isn't a capital "W"

The strings don't match because there's extra space before the quotation mark

△ Exactly the same

Strings must match for them to be equal. That means they must use capital letters, spaces, and symbols in exactly the same way.



EXPERT TIPS

Operator for strings

The "in" operator can be used to see whether one string is inside another string. Use it to check if a string contains a particular letter or a group of letters.

This checks whether "a" is in "abc"

```
>>> 'a' in 'abc'
```

True

```
>>> 'd' in 'abc'
```

False

"d" is not in "abc", so the answer is "False"